Processing of the Oil-Producing Well Testing Data by Means of Fuzzy *f-regression* Method

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Abstract. Present paper expands on the fuzzy input – fuzzy output regression analysis method that employs the degree of membership of a fuzzy point to a crisp parametric curve (hereafter "*f - regression*"). Method outline and formulae are provided.
A generalized "Maximum Plausiblity Estimator" concept is introduced as a formal means for defining the best fit of a parametric curve to fuzzy data concerning *à priori* uncertainty information. Clustering *f-regression* extension is defined for use in conjunction with "generalizing", "majority" and "grouping" estimators introduced in previous papers. An artificial example of clustering *f - regression* is provided.
Last section contains a well testing example of the pressure build-up analysis. The focus of this example is to show the *f - regression* capability to correctly identify linear fragments in the data even if each fragment contributes only a small portion of the ensemble.

Keywords: f-regression, fuzzy data, fuzzy plausibility, measure of plausibility, maximum plausibility estimate, data clustering, well testing, bottomhole pressure, build-up curve.

1 Introduction

An oilfield is a natural geological object containing an unknown quantity of hydrocarbons that represent a commercial opportunity. Modern oilfield development planning involves building 3D computer models based on the data from seismic surveys and exploration wells and updating them as more data becomes available. As drilled wells provide direct access to the field, they are perceived the most reliable data source. Extensive research is carried out to extract every bit of useful information from the (very limited) well data for the sake of oilfield evaluation.

Well testing is a means of obtaining crucial parameters of the near-wellbore zone: skin-factor, transmissibility, formation features etc. It involves producing a well at a known rate while recording the bottomhole pressure. In particular, "pressure build-up" test is conducted after the well is "shut-in" (rate = 0). Pressure build-up and pressure derivative profiles reflect multiple simultaneously occurring processes melded into a single curve. The effects of these processes on the resulting curve have been investigated to a great extent [1]. Matching pressure derivative in log-log coordinates with an ideal "type curve" is a common technique used by reservoir engineers. Linear fragments of the derivative profile indicate the domination of a particular effect, and getting parameter estimates by fitting groups of points with straight lines allows obtaining sought-after filtration parameters.

In **Section 5** we provide an example of the automatic discovery of linear features in the pressure build-up data leveraging the uncertainty information available.

2 Linear Regression Analysis – an Overview

General regression analysis problem is formulated as follows: based on a limited number N of observations of independent variables $\boldsymbol{x} \in \mathbf{R}^n$ and dependent variable y one has to identify such vector $\boldsymbol{a} \in \mathbf{R}^p$ that provides *the best* (in a strictly defined sense) prediction capabilities for $y = f(\boldsymbol{x}, \boldsymbol{a})$, where $f: \mathbf{R}^n \rightarrow \mathbf{R}^1$.

Linear least squares (LLS) criterion introduced by Karl Gauss uses the assumption that for each observation $i = \overline{1, N}$,

$$y_i = \boldsymbol{x}_i^T \cdot \boldsymbol{a} + \varepsilon, \tag{1}$$

with normally distributed measurement error $\varepsilon \sim N(0, \sigma)$. Hence the regression problem is reduced to finding such $\boldsymbol{a}$ that minimizes the dispersion of ε formulated as

$$D(\varepsilon) = \|\boldsymbol{y} - \boldsymbol{X} \cdot \boldsymbol{a}\|^2 \underset{\boldsymbol{a}}{\rightarrow} \min, \tag{2}$$

For well-defined experimental matrix $\boldsymbol{X}$, (2) is solved analytically, providing the equation that has the sense of conditional expectation of y: $E(y|\boldsymbol{x}) = f(\boldsymbol{x}, \boldsymbol{a})$.

The method has an inherent measure of both initial and residual uncertainty in a probabilistic sense, but no mechanism to account for *à priori* uncertainty information. A generalization of the LLS employs weights to reflect individual points' reliability, but no other inherent means are present.

Fuzzy regression analysis techniques, on the contrary, acknowledge the fact that "Uncertainty is an attribute of information" [2]. Most notably, **possibilistic regression** initially presented by Tanaka et al. [3] has the built-in capability to explicitly specify uncertainty information.

3 Fuzzy Linear *f-regression* Method

Definition 1. Fuzzy number $A = (\bar{a}, \alpha)_L$ defined on $U = \mathbf{R}^1$ is of *symmetric L-type* (or L-number for simplicity) if there are real numbers $\bar{a}$, $\alpha (> 0)$ and a function L with

1. L: $[0,\infty) \rightarrow [0,1]$; $L(0) = 1$, $L(\infty) = 0$.
2. L is continuous, strict monotonous decreasing and

$$\mu_A(t) = L\left(\frac{|t - \bar{a}|}{\alpha}\right). \tag{3}$$

Here, $\bar{a}$ is the *mean value* and α is the *spread* of A. (For $\alpha = 0$ we get the formal representation of crisp numbers). ■

In [4], a fuzzy regression approach was proposed to fit fuzzy input $X_{ji} = (\overline{x}_{ji}, \beta_{ji})_L$ and output $Y_i = (\overline{y}_i, \alpha_i)_L$ data with crisp linear parametric models. The following L-function is used in the implementation:

$$L(u) = e^{-u^m}, m > 1, \quad (4)$$

with m being the *form parameter,* allowing to emulate various types of fuzzy numbers from triangular to interval depending on the value of m chosen.

3.1 Measure of Similarity between a Fuzzy Point and a Parametric Curve

Definition 2 [5]**.** Let $\boldsymbol{x} \in \mathbf{R}^n, \boldsymbol{a} \in \mathbf{R}^p$, and be $y = f(\boldsymbol{x}, \boldsymbol{a})$ a crisp function. For given fuzzy point $\boldsymbol{Q}_i$ we introduce the *similarity measure* of $\boldsymbol{Q}_i$ with function f (for vector $\boldsymbol{a}$) by

$$M_i(\boldsymbol{a}) = \sup_{\boldsymbol{x} \in \mathbf{R}^n} \mu_{Q_i}(\boldsymbol{x}, f(\boldsymbol{x}, \boldsymbol{a})) \quad (5)$$

with $M_i(\boldsymbol{a}) = 1$ *or* $M_i(\boldsymbol{a}) = 0$ interpreted as "fuzzy point $\boldsymbol{Q}_i$ **belongs *or* does not belong** to the function f with vector $\boldsymbol{a}$". ■

There are two ways to formally introduce fuzzy points to *f-regression*.

1. **Production *T*-norm.** Assuming spreads for all input and output variables are independent, experimental data representation by fuzzy points $\boldsymbol{Q}_i = (\boldsymbol{X}_i, Y_i)$ is given as:

$$\mu_{Q_i}(\boldsymbol{x}, y) = \mu_{X_{1i}}(x_1) \cdot \ldots \cdot \mu_{X_{1n}}(x_n) \cdot \mu_{Y_i}(y) \quad (6)$$

Because of the choice of (4) and (6), for a linear model explicit analytical dependencies for the point's *similarity measure* $M_i(\boldsymbol{a})$ can be obtained:

$$M_i(\boldsymbol{a}) = L\left(\frac{|d_i(\boldsymbol{a})|}{s_i(\boldsymbol{a})}\right), \quad (7)$$

with $$d_i(\boldsymbol{a}) = a_0 + \sum_{j=1}^{n} a_j \cdot \overline{x}_{ji} - \overline{y}_i \quad (8)$$

and $$s_i(\boldsymbol{a}) = \left(\alpha_i^{\frac{m}{m-1}} + \sum_{j=1}^{n} |a_j \cdot \beta_{ji}|^{\frac{m}{m-1}} - \overline{y}_i\right)^{\frac{m-1}{m}} \quad (9)$$

2. **Elliptic fuzzy points.** We refer the reader to [6] for further information on the elliptic fuzzy points and to [7] for an *f-regression* case utilizing this approach.

As one can see, the *à priori* uncertainty information is captured in *the spreads* and *form attributes* of fuzzy numbers and transformed directly and explicitly into a measure of fitness between a curve and a fuzzy point.

3.2 Measure of Similarity between the Ensemble and a Parametric Curve

To propagate the described similarity metric through the dataset of N fuzzy points, the family of *mean-based aggregation operators* has been initially introduced in [4].

Here we generalize this idea through the concept of *plausibility*. For example, $M_i(\boldsymbol{a})$ provides the maximum plausibility of assumption "i-th fuzzy point in the ensemble belongs to $y = f(\boldsymbol{x}, \boldsymbol{a})$", which makes $M_i(\boldsymbol{a})$ the *maximum plausibility estimator* for given i, f and $\boldsymbol{a}$. Aggregated plausibility estimators (referred to as the "aggregation operators" in [4] and [7]) provide the following *measures of plausibility*:

1. **Geometric mean** provides plausibility estimate of the assertion "curve with parameters $\boldsymbol{a}$ fits all points in the ensemble simultaneously":
$$MG(\boldsymbol{a}) = \prod_i M_i^{w_i}(\boldsymbol{a}) \ \ s.t. \sum_i w_i = 1 \tag{10}$$
2. **Arithmetic mean** provides plausibility estimate of the assertion "curve with parameters $\boldsymbol{a}$ fits the majority of points in the ensemble":
$$MA(\boldsymbol{a}) = \sum_i w_i M_i(\boldsymbol{a}) \ \ s.t. \sum_i w_i = 1 \tag{11}$$
3. **Quadratic mean** provides plausibility estimate of the assertion "curve with $\boldsymbol{a}$ fits the tight group of points in the ensemble":
$$MQ(\boldsymbol{a}) = \sqrt{\sum_i w_i M_i^2(\boldsymbol{a})} \ \ s.t. \sum_i w_i = 1 \tag{12}$$

The following definition formally introduces the "best fit" criteria of *f-regression*.

Definition 3. The *Maximum Plausibility Estimate* (MPE) is the vector of crisp linear model parameters $\boldsymbol{a}_{MP}$ that, for the given ensemble of fuzzy points, delivers maximum to the selected *plausibility measure* $MP(\boldsymbol{a})$:
$$\boldsymbol{a}_{MP} = \arg\max_{\boldsymbol{a}\in \mathrm{R}^p} MP(\boldsymbol{a}). \tag{13}$$
NB: As the form parameter *m* impacts the MPE for the given MP, the value should be indicated for disambiguation where necessary, e.g. $\boldsymbol{a}_{MQ}^{m=4}$. ■

3.3 Data Clustering

Initial approach to data clustering via *f-regression* was discussed in [7]. Here we provide a generalized approach to the regression analysis of experimental data representing multiple classes, each described by a linear model with unique parameters $\boldsymbol{a}^{(1)}, \boldsymbol{a}^{(2)}$ etc. The plausibility measure for the assumption "i-th point belongs to **any** class", $M_i^C(\dots)$, extends the ordinary similarity measure via:
$$M_i^C(\boldsymbol{a}^{(1)}, \dots, \boldsymbol{a}^{(k)}) = M_i(\boldsymbol{a}^{(1)}) \tilde{\mp} \dots \tilde{\mp} M_i(\boldsymbol{a}^{(k)}) \tag{14}$$
where k is the total number of classes, chosen by the experimenter, and $\tilde{\mp}$ denotes the **algebraic sum** *S*-norm.

Any aggregated plausibility estimator MP (10)-(12) may be used in conjunction with (14). In general, upon maximization every MP provides unique parameter vectors that are "Maximum Plausibility Estimates" in their particular sense.

Section 4 provides an example and contains some practical considerations.

4 Impact of Data Uncertainty on the Model Fitness

Consider the following artificial example (Fig. 1). We have taken 7 points belonging to $y = 2{,}5 + 1{,}5 \cdot x$ (*Class 1*) and injected points #5-#7 belonging to $y = 7{,}5 + 1{,}2 \cdot x$ (*Class 2*). Both dependent and independent variable were added uniformly distributed noise $\varepsilon_x = \mathrm{U}(-0.25, 0.25)$ and $\varepsilon_y = \mathrm{U}(-0.5, 0.5)$ with $r_{xy} = 0$ to obtain $\overline{x} = x + \varepsilon_x$ and $\overline{y} = y + \varepsilon_y$ that were given to the LLS and *f-regression* to obtain parameter estimates (Table 1). This section provides an example of how initial dataset can be reconstructed from the uncertainty attributes.

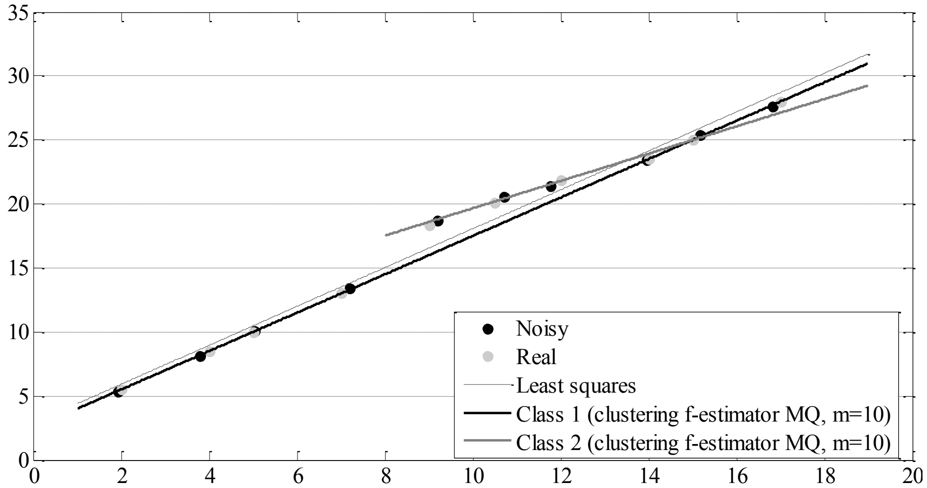

Fig. 1. LLS and Clustering f-regression fitting experimental data

First, the least-squares regression estimates (dashed line) are obtained as a benchmark. Residual dispersion $D_{y|x} = 8.04$ $(\sigma_{y|x} = 2.84)$. Since no *à priori* uncertainty attributes are utilized by LLS, the same estimate can be obtained with *f-regression* via $MP = MG$ (geometric mean): if we set $m = 2, \beta_{ji} = 0, \alpha_i =$ any positive number $\forall i$, we obtain **exactly** the same result. As one can see from (8)-(10), maximizing (11) is numerically equivalent to minimizing (3), which enables *f-regression* to produce LLS estimates $\boldsymbol{a}_{LS}$ as a special case.

Second, we explicitly model the introduced uniform noise through fuzzy numbers with $m = 10, \beta_i = 0.25, \alpha_i = 0.5$ $\forall i$ selecting $MP = MQ$ (quadratic mean) for group detection. Cross-check of LLS estimates $\boldsymbol{a}_{LS}$ against these uncertainty assumptions reveals already quite high plausibility (0.817) for $\boldsymbol{a}_{LS}$. Furthermore, *f-regression* estimated parameters $\boldsymbol{a}_{MQ}^{m=10}$ are almost indistinguishable from the LLS ones (see Table 1, under *f-estimator*) and therefore not present on the graph. Point-by-point comparison shows that *f-estimates* have excellent fit across *Class 1* points and $M_7(\boldsymbol{a}) = 0.652$ (points #5 and #6 clearly marked as "outliers"). The *f-estimates* get biased towards point #7 to maximize plausibility (0.844) due to near-interval uncertainty.

(***NB:*** It is of practical relevance that running the same dataset with form parameter $m = 4$ provides $a_{MQ}^{m=4} = [2.49, 1.50]$ that are spot-on for *Class 1*, and marks all three *Class 2* points as "outliers". Collected empirical evidence suggests $m = 4$ as the "best bet" for many problems.)

Independent results show that the difference between conventional regression and *f-regression* can be insignificant even for carefully picked uncertainty data [8]. Running into this situation might be just the nature of the problem under consideration, or might indicate excessive uncertainty in the data.

Third, we want to see if outlying points form a specific class. Therefore clustering *f-regression* is set with $k = 2$ classes, with all other parameters left intact.

Table 1. Source data and results for the artificial problem in Section 4

	Real data		"Noisy" data		LLS	*f-estimator* $MP = MQ$		Clustering *f-estimator* $MP = MQ$ (using M_i^C)		
#	x	y	$\overline{x}$	$\overline{y}$	$M_i(\boldsymbol{a})$ ($m = 10$)	$M_i(\boldsymbol{a})$ ($m = 10$)	$M_i(\boldsymbol{a})$ ($m = 4$)	$M_i^C(\dots)$	$M_i(\boldsymbol{a}^{(1)})$	$M_i(\boldsymbol{a}^{(2)})$
1	2.0	5.5	1.94	5.38	0.9980	0.9927	1	1	1	0
2	4.0	8.5	3.80	8.10	0.9825	0.9677	0.9997	1	1	0
3	5.0	10.0	5.05	10.10	0.9976	0.9965	1	1	1	0
4	7.0	13.0	7.20	13.40	0.9989	0.9992	0.9996	1	1	0
5	9.0	18.3	9.19	18.68	0	0	0	1	0	1
6	10.5	20.1	10.73	20.56	0	0	0	1	0	1
7	12.0	21.9	11.76	21.41	0.8700	0.6517	0.0001	1	0	1
8	14.0	23.5	13.95	23.41	0.8787	0.9770	1	1	1	0.9865
9	15.0	25.0	15.18	25.36	0.9699	0.9972	0.9998	1	1	1
10	17.0	28.0	16.82	27.64	0.4976	0.9114	0.9997	1	1	0.4361
					a_{LS}	$a_{MQ}^{m=10}$	$a_{MQ}^{m=4}$	–	$a_{MQ}^{(1)}$	$a_{MQ}^{(2)}$
				$a_0 =$	2.8750	2.9605	2.4926		2.4963	8.9875
				$a_1 =$	1.5195	1.5056	1.5007		1.5004	1.0669
Residual dispersion (LLS)					**8.04**	**8.11**	**11.27**	–	–	–
Plausibility measure (*f-regression*)					**0.8168**	**0.8437**	**0.8365**	**1**	**0.8367**	**0.7186**

Further observations can be made to highlight the results (Fig. 1):

- Plausibility measure for $MQ^C(\boldsymbol{a}^{(1)}, \boldsymbol{a}^{(2)}) = 1$ (see Table 1, under "Clustering *f-estimator*"), meaning no points are left outside either class.
- Estimates $a_{MQ}^{(1)} = [2.50, 1.50]$ are spot-on (thick black line). Comparing previously obtained *MQ* ($m = 10$) with *Class 1*, one can see that $M_7(\boldsymbol{a}) = 0.652$ vs. $M_7(\boldsymbol{a}^{(1)}) = 0$, the decrease enabled and compensated by $M_7(\boldsymbol{a}^{(2)}) = 1$.
- Estimates $a_{MQ}^{(2)} = [8.99, 1.07]$ are noticeably different from their true values (thick grey line), explained by too few *Class 2* points in the ensemble.
- Estimator MQ^C is highly non-linear with plethora of local extrema. Optimization method used here is genetic algorithm (MATLAB implementation). Fine-tuning is required in order to reliably obtain correct estimates.

5 Well-Testing Data Analysis

Pressure buildup test is one of the key methods used throughout the reservoir development, and it is conducted as follows. After an oil well has been producing at a rate q for t_p hours, it is closed ("shut-in"). The bottomhole pressure difference, $\Delta p = p(t) - p(t_p)$, is measured vs. the time since the shut-in, $\Delta t = t - t_p$. The plots of $\Delta p(\Delta t)$ and its derivative are known to exhibit linear behavior in log-log or semi-log coordinates, based on the currently dominating effect such as e.g. wellbore storage ("afterflow") and skin in early time, heterogeneous formation behavior in mid-time and homogenized formation radial flow in late time [1].

Example used in this section is a dual-porosity pressure buildup test Example 2 presented in [9]. The following combined plot shows pressure build-up and pressure change rate over "effective time", $t_e = \frac{t_p \cdot \Delta t}{t_p + \Delta t}$, converted into dimensionless units, p_D vs. t_D / C_D, on a log-log scale (Fig. 2).

5.1 Assumptions and Results

The following key assumptions were used in the analysis:

- **Form parameter** has been set to $m = 4$.
- **Number of classes** (linear fragments) has been set to $k = 4$.
- **Weights** of data points across the ensemble were adjusted based on their perceived "density": log-log $\Delta p(\Delta t)$ curve clearly has accumulations of data points in certain areas, whereas they provide little additional information to the observer; hence, their impact should be lowered. Therefore the weights w_i of experimental points were adjusted based on the density δ_i of data around the i-th point (with ω defining the width of proximity "window") as

$$w_i = \frac{\delta_i^{-1}}{\sum_j \delta_j^{-1}}, \text{ where } \delta_i = \sum_{j=1}^{N} L\left(\frac{\left|\log_{10}\left(\frac{t_D}{C_D}\right)_i - \log_{10}\left(\frac{t_D}{C_D}\right)_j\right|}{\omega}\right). \quad (15)$$

- **Time-based data** is a difference between the time measured with discretization of ca. 1.85 seconds and shut-in time, resulting in the base spread of 0.92 seconds. Furthermore we adjusted individual spreads based on the value of Δt to accommodate rounding errors from having only 5 significant digits.
- **Pressure data** uncertainty stems from (1) measurement errors (0.01 bar) in both $p(t)$ and $p(t_p)$ and value-dependent rounding error in Δp; (2) multiple phenomena producing stochastic behavior. Varying spreads were adopted in early, mid and late times, averaging 1% of mean value across the board.
- **Pressure derivative data** uncertainty is derived from the source variables. The weighted formula used in [9] for the derivative calculation at point i is:

$$\left.\frac{\mathrm{d}p}{\mathrm{d}F}\right|_i = \frac{\frac{p_i - p_{i-1}}{F_i - F_{i-1}}(F_{i+1} - F_i) + \frac{p_{i+1} - p_i}{F_{i+1} - F_i}(F_i - F_{i-1})}{F_{i+1} - F_{i-1}} \quad (16)$$

  with $F = \ln t_e$, denoting the time function used.

As described in [7], relations involving L-numbers can be approximated using first-order Taylor series expansion that uses only unambiguous operations: addition and multiplication by a constant. Applying a sequence of transformations, we can estimate the spread of pressure derivative for each point.

While efficiently smoothing the p'_D curve, applying (16) leads to the escalation of fuzziness, yielding the average spread of pressure derivative amounting to 163% of the mean value. Therefore it has been substituted by the uncertainty of a simplified non-weighted formula, averaging to 15% of the absolute derivative value.

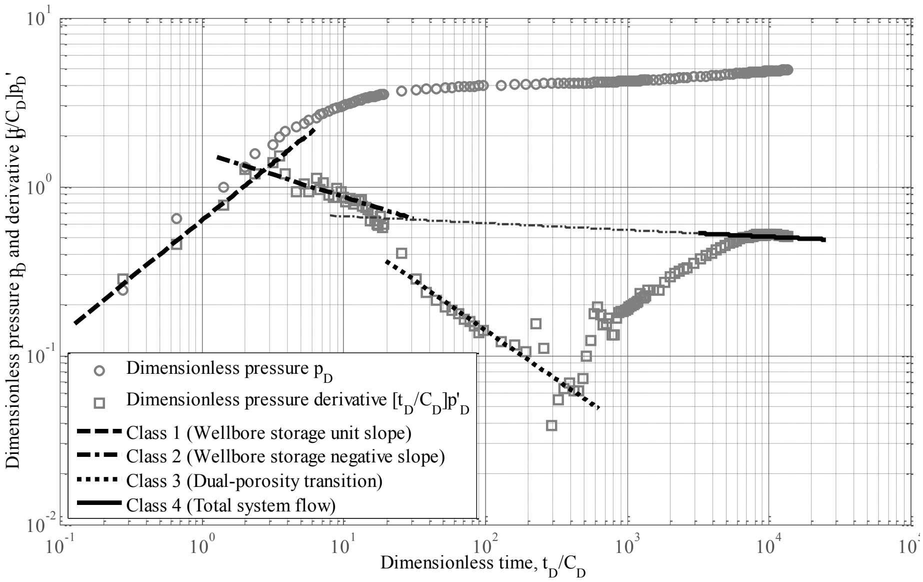

Fig. 2. Log-log match of dual-porosity data with linear fragments

Table 2. Results of the clustering *f-regression* applied to pressure build-up data

Classes	MPEs $a_{MQ}^{(i)}$ $p'_D = 10^{a_0}\left[\frac{t_D}{C_D}\right]^{a_1}$		Number of fitting points ($M_i(\boldsymbol{a})$>0.9)		Number of outliers ($M_i(\boldsymbol{a})$<0.1)		$MQ(\boldsymbol{a})$
	a_0	a_1	Count	Weighted	Count	Weighted	
Class 1	–0.1983	0.6797	6	17.5	144	130.1	0.3413
Class 2	0.2010	–0.2617	21	23.5	113	109.6	0.4440
Class 3	0.3192	–0.5839	22	47.5	126	98.2	0.5639
Class 4	–0.1348	–0.0403	48	18.3	97	128.9	0.3539
All classes			**94**	**94.1**	**37**	**38.3**	**0.8150**

151 data points were fitted with 4 linear models, resulting in an 8-dimensional optimization problem. Solving the problem provided plausibility and MPE of the clustering *f-regression* shown in Table 2. Further comments highlight the obtained results.

Early time ($t_D/C_D < 12$) consists of *(1)* **wellbore storage unit slope** described by *Class 1* ($a_1^{(1)} \neq 1$ suggests that the shut-in time can be fine-tuned), *(2)* maximum arch, *(3)* **wellbore storage negative slope** described by *Class 2* (see note at the end of this section) and *(4)* transition radial period (virtually non-existent in this example).

Mid-time ($t_D/C_D < 5000$) shows characteristic **dual-porosity transition** valley with $p_D' < 0.5$. Initial homogenous behavior of high-conductivity fissures is now complemented by the response of porous matrix. *Class 3* parameters allow further estimation of e.g. storativity ratio and interporosity flow.

Late time ($t_D/C_D > 5000$) shows the homogenized total system radial flow with theoretical $p_D' = 0.5$ (zero slope). Descending *Class 4* slope $a_1^{(4)} = -0.04$ suggests the well system is gradually approaching reservoir pressure.

NB: Two observations should be made. First, ensemble-wide generalization of the clustering *f-regression* can lead to biased estimates (see dotted extension of the *Class 4* line interfering with *Class 2*). Second, there is a lack of smooth transition between the fragments. Piecewise-linear or Sugeno extensions can be introduced to *f-regression* to further improve the results.

MATLAB scripts for running *f-regression* and the examples in this paper can be downloaded from http://bizyumov.gubkin.ru/fest/matlab/.

6 Conclusions

Generalized *measure of plausibility* (MP) concept is defined as a formal means of evaluating model fitness provided *à priori* uncertainty attributes. Maximizing MP yields the *maximum plausibility estimate* (MPE) of parameter values. An extension of the previously published *f-regression* method is discussed that allows fitting the ensemble of fuzzy points with multiple linear fragments.

A well testing analysis example is provided that shows the capability of *f-regression* to reproduce the approach of domain expert and accurately identify multiple linear trends on the log-log scale. The paper demonstrates how the explicit *à priori* uncertainty information can be utilized to identify complex non-linear datasets reflecting multiple simultaneous processes merged into one curve.

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